lobster's shell is obviously a part of its body. But what about the caddis house?

Natural selection favoured those ancestral caddis genes that caused their possessors to build effective houses. The genes worked on behaviour, presumably by influencing the embryonic development of the nervous system. But what a genticist would actually see is the effect of genes on the shape and other properties of houses. The geneticist should recognize genes 'for' house shape in precisely the same sense as there are genes for, say, leg shape. Admittedly, nobody has actually studied the genetics of caddis houses. To do so you would have to keep careful pedigree records of caddises bred in captivity, and breeding them is difficult. But you don't have to study genetics to be sure that there are, or at least once were, genes influencing differences between caddis houses. All you need is good reason to believe that the caddis house is a Darwinian adaptation. In that case there must have been genes controlling variation in caddis houses, for selection cannot produce adaptations unless there are hereditary differences among which to select.

Although geneticists may think it an odd idea, it is therefore sensible for us to speak of genes 'for' stone shape, stone size, stone hardness, and so on. Any geneticist who objects to this language must, to be consistent, object to speaking of genes for eye colour, genes for wrinkling in peas, and so on. One reason the idea might seem odd in the case of stones is that stones are not living material. Moreover, the influence of genes upon stone properties seems especially indirect. A geneticist might wish to claim that the direct influence of the genes is upon the nervous system that mediates the stone-choosing behaviour, not upon the stones themselves. But I invite such a geneticist to look carefully at what it can ever mean to speak of genes exerting an influence on a nervous system. All that genes can really influence directly is protein synthesis. A gene's influence upon a nervous system, or, for that matter, upon the colour of an eye or the wrinkliness of a pea, is *always* indirect. The gene determines a protein sequence that influences *X* that influences *Y* that influences *Z* that eventually influences the wrinkliness of the seed or the cellular wiring up of the nervous system. The caddis house is only a further extension of this kind of sequence. Stone hardness is an *extended* phenotypic effect of the caddis's genes. If it is legitimate to speak of a gene as affecting the wrinkliness of a pea or the nervous system of an